

Homomorphisms of structures in simplicial type theory

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Introduction

- $\text{HoTT} = \text{MLTT} + \text{UA} + \text{HITs}$
- $\text{UA} \rightarrow \text{SIP} = \text{“identity types are equivalent to isomorphisms”}$
[CD13; Uni13; Ahr+25]

Example

SIP for Monoids:

$$(M, \cdot_M, \varepsilon_M) = (N, \cdot_N, \varepsilon_N) \simeq \text{Iso}((M, \cdot_M, \varepsilon_M), (N, \cdot_N, \varepsilon_N))$$

SIP with Directions?

SIP is fundamentally **directionless**: the identity type is symmetric

¹LH11; War13; Nuy15; RS17; Nor18; Kav19; Nuy20; WL20; ANW23; Wea24; NA24; Neu25.

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Question

Is it possible to find a directed analogue to the SIP? A structure homomorphism principle [WL20] that would similarly characterise homomorphisms of structures?

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Yes! Directed identity type¹ + directed univalence

Define the type of monoids

Directed identity type = monoid homomorphisms

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Simplicial Type Theory

$$\text{STT} = \text{HoTT} + \text{Interval}$$

Simplicial Type Theory

STT = HoTT + Interval

Interval : a bounded totally ordered set with points $0 : \mathbb{I}$ and $1 : \mathbb{I}$ [RS17]

$$0 \longrightarrow 1$$

$$\mathrm{hom}_A(a, b) \triangleq \sum_{p: \mathbb{I} \rightarrow A} p(0) = a \times p(1) = b$$

The Behaviour of the hom type

Question

Does the morphism type make it an ∞ -category?

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No!

The Behaviour of the `hom` type

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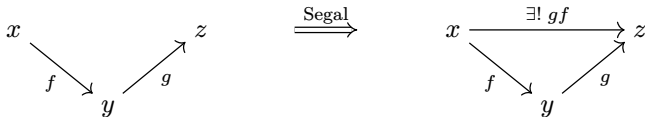
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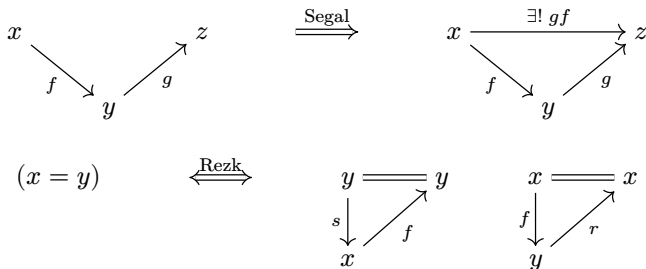


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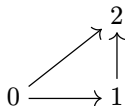
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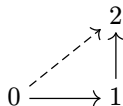
Composition Problems

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$$\Delta^2 \triangleq \sum_{r,s:\mathbb{I}} r \leq s$$

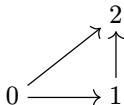

 Δ^2

$$\Lambda_1^2 \triangleq \sum_{r,s:\mathbb{I}} (r = 0) \vee (s = 1)$$

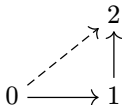

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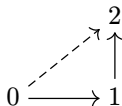
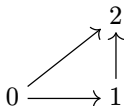
$\Lambda_1^2 \rightarrow X \leftrightarrow$ a pair of composable homs in X

$\Delta^2 \rightarrow X \leftrightarrow$ a commutative triangle in X

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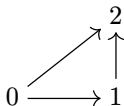
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$X^{\Delta^2} \rightarrow X^{\Lambda_1^2}$ restriction map

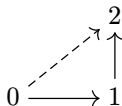
$X^{\Delta^2} \xrightarrow{\sim} X^{\Lambda_1^2}$ then homs in X compose

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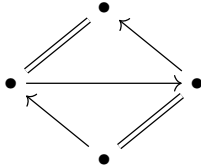
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$X^{\Delta^2} \xrightarrow{\sim} X^{\Lambda_1^2}$ then homs in X compose or X is Segal

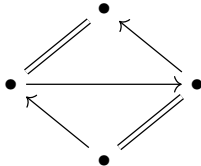
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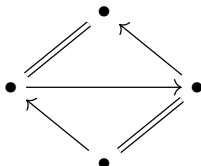


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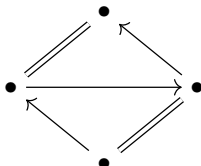
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WARNINGS ⚠

- $\mathcal{S} \subseteq \mathcal{U}$ and $\mathcal{S}_{\leq 0} \subseteq \mathcal{U}_{\text{isSet}}$
- $b(\mathcal{S}_{\leq 0}) = b(\mathcal{U}_{\text{isSet}}) \leftarrow$ same objects different morphisms
- $\mathcal{U}_{\text{isSet}}$ is **not** a synthetic category

Directed Univalence for Spaces

Theorem

Synthetic morphisms between spaces $\text{hom}_g(A, B)$ are equivalent to ordinary functions $A \rightarrow B$

$$\text{dua} : \forall (A, B : \mathcal{S}) \rightarrow \text{hom}(A, B) \simeq (A \rightarrow B)$$

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HoTT	STT
$\mathcal{U}$	$\mathcal{S}$
$\mathcal{U}_{\text{isSet}}$	$\mathcal{S}_{\leq 0}$
univalence	directed univalence

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Problem?

$\mathcal{U}_{\text{isSet}}$ is not a synthetic category **✗**

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Replace $\mathcal{U}_{\text{isSet}}$ with $\mathcal{S}_{\leq 0}$ **✓**

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Homomorphisms of Structures

Σ -types commute with the identity types: a path of (dependent) pairs is a pair of (dependent) paths

$$\begin{array}{ccc}
 \langle a : A, b : B(a) \rangle & & \text{transport}_p^B(b) \\
 \parallel & \parallel_p & \parallel \\
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Synthetic morphisms of Σ -types are pairs of synthetic morphisms

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 f \downarrow & \downarrow_p & \downarrow \\
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(assuming B is a covariant family)

The Structure Homomorphism Principle

$$\text{magmaSHP} : \text{hom}((M, \cdot_M), (N, \cdot_N)) \simeq \sum_{f: M \rightarrow N} (\cdot_N) \circ (f \times f) = f \circ (\cdot_M)$$

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A synthetic morphism of magmas is a function $f : M \rightarrow N$ such that the diagram

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SHP for Algebraic Theories

Type	Morphisms
Magma	magma homomorphisms
Monoid	monoid homomorphisms
Ring	ring homomorphisms
models of an algebraic theory T in $\mathcal{S}_{\leq 0}$	T -algebra homomorphisms

Limited to algebraic theories?

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No!

$$\text{AnalyticPecat} \triangleq \sum_{\text{ob}:\mathcal{S}_{\leq 0}} \sum_{\text{mor}:\text{ob} \times \text{ob} \rightarrow \mathcal{S}_{\leq 0}} \sum_{\text{id}:\prod_{x:\text{ob}} \text{mor}(x,x)} \dots$$

Theorem

AnalyticPecat forms a synthetic category whose objects are analytic precategories synthetic morphisms are equivalent to functors

Conclusions and Future Work

- Establish SHPs for algebraic theories
- ... and for precategories
- A lot of **synthetic mathematics** on the way!
- Formalisation in Cubical Agda [VMA21] building up on ideas from de Jong, Kraus, and Ljungström [dKL26]

Thank you!

- [Ahr+25] Benedikt Ahrens et al. *The Univalence Principle*. Memoirs of the AMS. American Mathematical Society, 2025. ISBN: 978-1-4704-7269-6.
- [ANW23] Benedikt Ahrens, Paige Randall North, and Niels van der Weide. “Bicategorical type theory: semantics and syntax”. In: *Mathematical Structures in Computer Science* 33.10 (Oct. 2023), pp. 868–912. ISSN: 1469-8072. DOI: 10.1017/s0960129523000312. URL: <http://dx.doi.org/10.1017/S0960129523000312>.
- [CD13] Thierry Coquand and Nils Anders Danielsson. “Isomorphism is equality”. In: *Indagationes Mathematicae* 24.4 (Nov. 2013), pp. 1105–1120. ISSN: 0019-3577. DOI: 10.1016/j.indag.2013.09.002.
- [dKL26] Tom de Jong, Nicolai Kraus, and Axel Ljungström. “The Leibniz adjunction in homotopy type theory, with an application to simplicial type theory”. In: *ArXiv abs/2601.21843* (2026). URL: <https://api.semanticscholar.org/CorpusID:285139886>.

- [GWB24] Daniel Gratzer, Jonathan Weinberger, and Ulrik Buchholtz. *Directed univalence in simplicial homotopy type theory*. 2024. arXiv: 2407.09146 [cs.LO]. URL: <https://arxiv.org/abs/2407.09146>.
- [Kav19] G.A. Kavvos. *A quantum of direction*. Online. 2019. URL: <https://seis.bristol.ac.uk/~tz20861/papers/meio.pdf>.
- [LH11] Daniel R. Licata and Robert Harper. “2-Dimensional Directed Type Theory”. In: *Electronic Notes in Theoretical Computer Science* 276 (Sept. 2011), pp. 263–289. ISSN: 1571-0661. DOI: 10.1016/j.entcs.2011.09.026. URL: <http://dx.doi.org/10.1016/j.entcs.2011.09.026>.
- [NA24] Jacob Neumann and Thorsten Altenkirch. *The Category Interpretation of Directed Type Theory*. Online. 2024. URL: <https://jacobneu.github.io/research/preprints/catModel-2024.pdf>.
- [Neu25] Jacob Neumann. “A Generalized Algebraic Theory of Directed Equality”. PhD thesis. University of Nottingham, 2025.

- [Nor18] Paige Randall North. *Towards a directed homotopy type theory*. 2018. arXiv: 1807.10566 [cs.LG]. URL: <https://arxiv.org/abs/1807.10566>.
- [Nuy15] Andreas Nuyts. “Towards a Directed Homotopy Type Theory based on 4 Kinds of Variance”. *MA thesis*. KU Leuven, 2015.
- [Nuy20] Andreas Nuyts. *A Vision for Natural Type Theory*. 2020. URL: <https://anuyts.github.io/files/nattt-vision.pdf>.
- [RS17] Emily Riehl and Michael Shulman. “A type theory for synthetic ∞ -categories”. In: *Higher Structures* 1 (1 2017), pp. 147–224. DOI: 10.21136/HS.2017.06.
- [Uni13] The Univalent Foundations Program. *Homotopy Type Theory: Univalent Foundations of Mathematics*. Institute for Advanced Study, 2013. URL: <https://homotopytypetheory.org/book>.
- [VMA21] Andrea Vezzosi, Anders Mörtberg, and Andreas Abel. “Cubical Agda: A dependently typed programming language with univalence and higher inductive types”. In: *Journal of Functional Programming* 31 (2021). ISSN: 1469-7653. DOI: 10.1017/s0956796821000034.

- [War13] Michael Warren. *Directed type theory*. Online. Seminar talk. Apr. 2013. URL: <https://www.ias.edu/video/univalent/1213/0410-MichaelWarren>.
- [Wea24] Matthew Zachary Weaver. “Bicubical Directed Type Theory”. PhD thesis. Princeton University, 2024. URL: <http://arks.princeton.edu/ark:/88435/dsp017s75dg778>.
- [WL20] Matthew Z. Weaver and Daniel R. Licata. “A Constructive Model of Directed Univalence in Bicubical Sets”. In: *Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science*. LICS '20. ACM, July 2020. DOI: 10.1145/3373718.3394794.